6 **(a)** Fig. 6.1 shows two resistors **X** and **Y** connected in series to a battery with an internal resistance. The voltmeter has an infinite resistance.

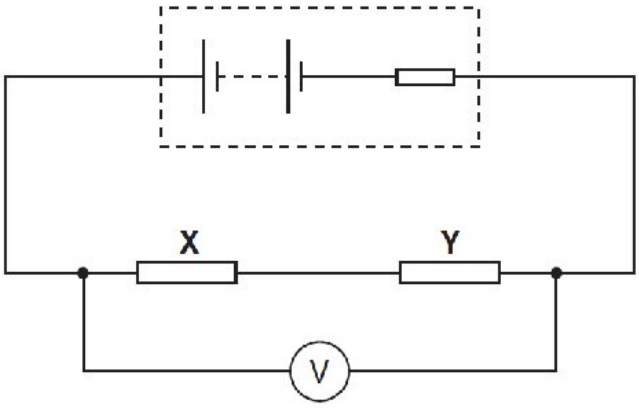


Fig. 6.1

(i) Explain why the voltmeter reading is not equal to the e.m.f. of the battery.

.....

..... [1]

(ii) The resistance of resistor **X** is four times the resistance of resistor **Y**. The voltmeter reading is 6.0 V.

Calculate the potential difference across the resistor **Y**.

potential difference = ... V [2]

(b) Fig. 6.2 shows four capacitors, each of capacitance 50 μF, connected to a high-voltage supply, a resistor and a switch S.

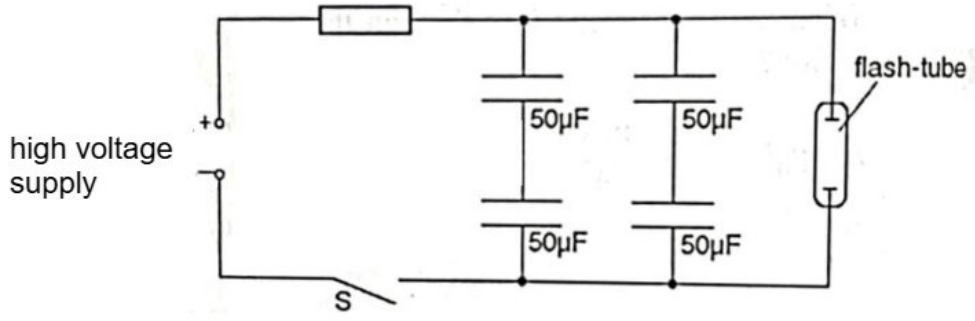


Fig. 6.2

A flash-tube is connected across the combination of capacitors. When S is closed, the capacitors charge up gradually and, when the potential difference across the flash-tube reaches 540 V, the capacitors discharge rapidly through the flash-tube.

A flash of light is emitted as the capacitors discharge 63% of their stored energy.

(i) Determine the total capacitance of the arrangement of capacitors.

capacitance = ... μF [2]

(ii) Calculate

1. the energy dissipated in one flash of light,

energy = ... J [2]

2. the potential difference across each capacitor immediately after the flash of light has occurred.

potential difference = ... V [2]

[Total: 9]